DATE 22/10/26 03

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Fundamentals:

Bits ⇒ 0s & 1s ⇒ Electrical signals

↓

Binary digits

With the two binary Digits together, we can create different patterns.

Ex) 2 binary digits ⇒ $2^2 = 4$

00, 01, 10, 11

3 binary digits ⇒ 000, 001, 010, 100, 011, 111, 110, 101

$2^3 \Rightarrow 8$.

Likewise:

Bit	Possible Patterns
1	2
2	4
4	16
8	256
32	4.3 billion
64	18.4 quintillion

DATE / / S M T W T F S

Bytes:

8 bits grouped together to form 1 byte

Computer Memory & file sizes are commonly measured in bytes:

- 1 byte ⇒ 8 bits
- 1 Kb ⇒ 1000 bytes
- 1 MB ⇒ 1 Million bytes
- 1 GB ⇒ 1 billion bytes.

Numeric Representation:

Say, we have 4 digit bit Representation:
Each will denote a Number

Ex) 0000 ⇒ 0
0001 ⇒ 1
0010 ⇒ 2
⋮
1111 ⇒ 15

Therefore, 4 bits ↓ (0-15) = 16 patterns

Likewise:
8 bits ⇒ 256 patterns
32 bits ⇒ 4.3 billion patterns.

DATE / /

S M T W T F S
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How Binary counting Works?

Ex) 0 1 0 0

8 4 2 1 ⇐ values of each bit position

(2^3) (2^2) (2^1) (2^0)

0 ⇒ Nothing

1 ⇒ Value taken into account.

∴ 0100 ⇒ 0 + 4 + 0 + 0 (only 4 has 1)

⇒ 4

∴ 0100 Represents 4

Another ex) 0011

0011 ⇒ 0 + 0 + 2 + 1

⇒ 3

∴ 0011 = 3

(Likewise for 8 digits)

Ex) 0 1 0 0 1 0 1 1 ⇒ 1 + 2 + 8 + 64

128 64 32 16 8 4 2 1 = 75

DATE / /

S M T W T F S
O O O O O O O

HexaDecimal ⇒ (Base 16)

Used for Shorter Representation of a Bit pattern.

∴ Hexa decimal symbol ⇒ Represents 4 bit values

From, 0 - 9 ⇒ Normal Decimal Values
From, 10 - 15 ⇒ A, B, C, D, E, F.

Now, 1111 0000 1111 ⇒ F0F

(2 1 0)

Now, Let's calculate it's value.

	Position	Calc (Base 16)	Value
F	2	15×16^2	3840
0	1	0×16^1	0
F	0	15×16^0	15
			16.

∴ Therefore, how many digits the number could be, we can split into 4 digits, give hexadecimal symbol with base 16. Hence, we can calculate it's value.

DATE / /

S M T W T F S
O O O O O O O

Convert Binary to Hexadecimal

1010 1100 ⇒ 10 10 ⇒ 8 + 2
⇒ 10

10 ⇒ A
12 ⇒ C

1 1 0 0 ⇒ 8 + 4 = 12
8 4 2 1

∴ AC is Hexadecimal

Value:

	Pos.	Value	Value
A	1	10×16^1	160
C	0	12×16^0	12
			172